

Multiscale Stigmergic Field — OWT 125M Training Report

Architecture

- 125M parameter GPT with chunk-parallel stigmergic attention
- Dual-scale social field: fast (fixed evap) + slow (learnable retention)
- Warm-started from 60K-step single-field checkpoint on OpenWebText

Key Findings at Step 29,000 / 70,000

1. PPL surpassed single-field baseline

Multiscale PPL 27.3 vs single-field best 27.4 — and still improving

2. Slow field is actively contributing

At step 25K, removing slow field hurts chunks 3-7 (tokens 384-1024)

Slow field accounts for 37-97% of total field effect in later chunks

3. Dramatic layer specialization

Layer 0 slow field: half-life 253 tokens (long-range memory)

Layers 1-11: half-life collapsed to 1-5 tokens (local processing)

Model self-organized: one long-memory layer + fast processing layers

4. Field effect grows with sequence position

Ablation delta: 0.0 at chunk 0 → +0.009 at chunk 7

Confirms field carries useful context forward across chunks

5. Retention evolves during training

Layer 0 half-life: 200 → 261 tokens over 29K steps (still growing)

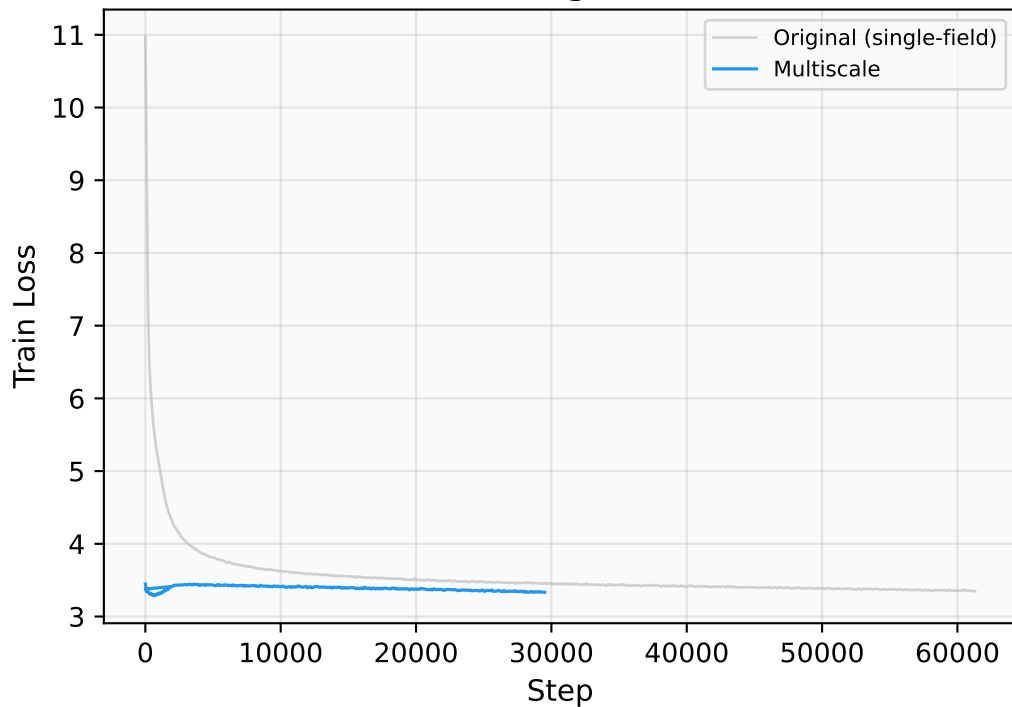
Deeper layers collapsed from 200 → 1 token (learned to be local)

Training Status

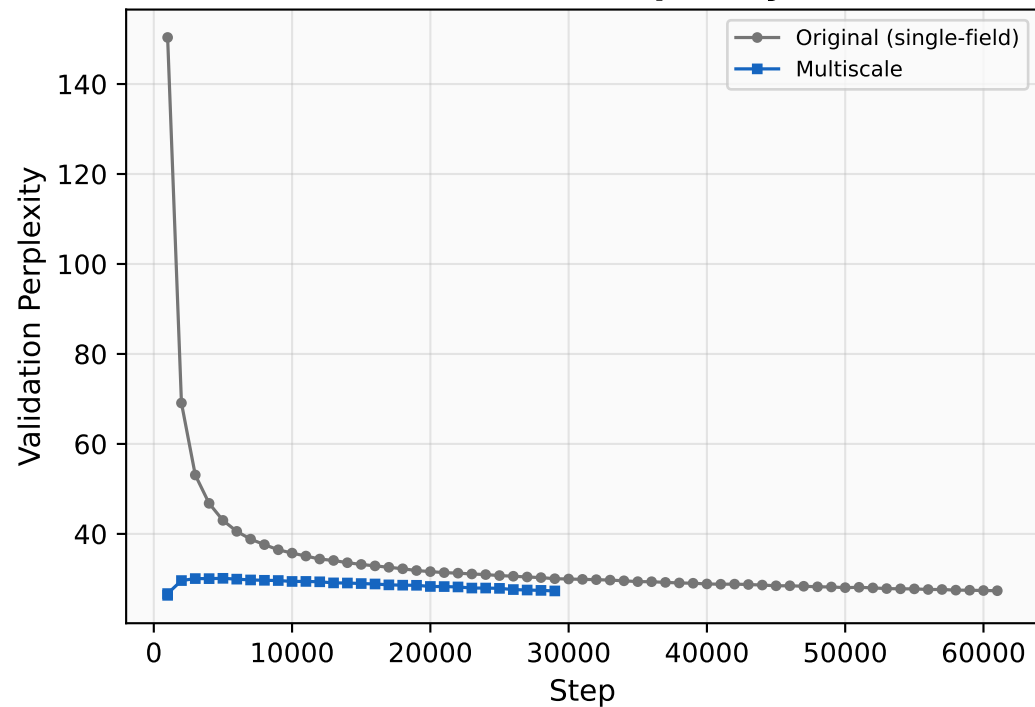
- Step 29,000 / 70,000 (41% complete)
- Throughput: 74K tokens/sec on L40S
- ETA: ~6.5 hours remaining
- LR in cosine decay phase ($4.1e-4$, peak was $6e-4$)

Multiscale vs Single-Field Training on OpenWebText (125M params)

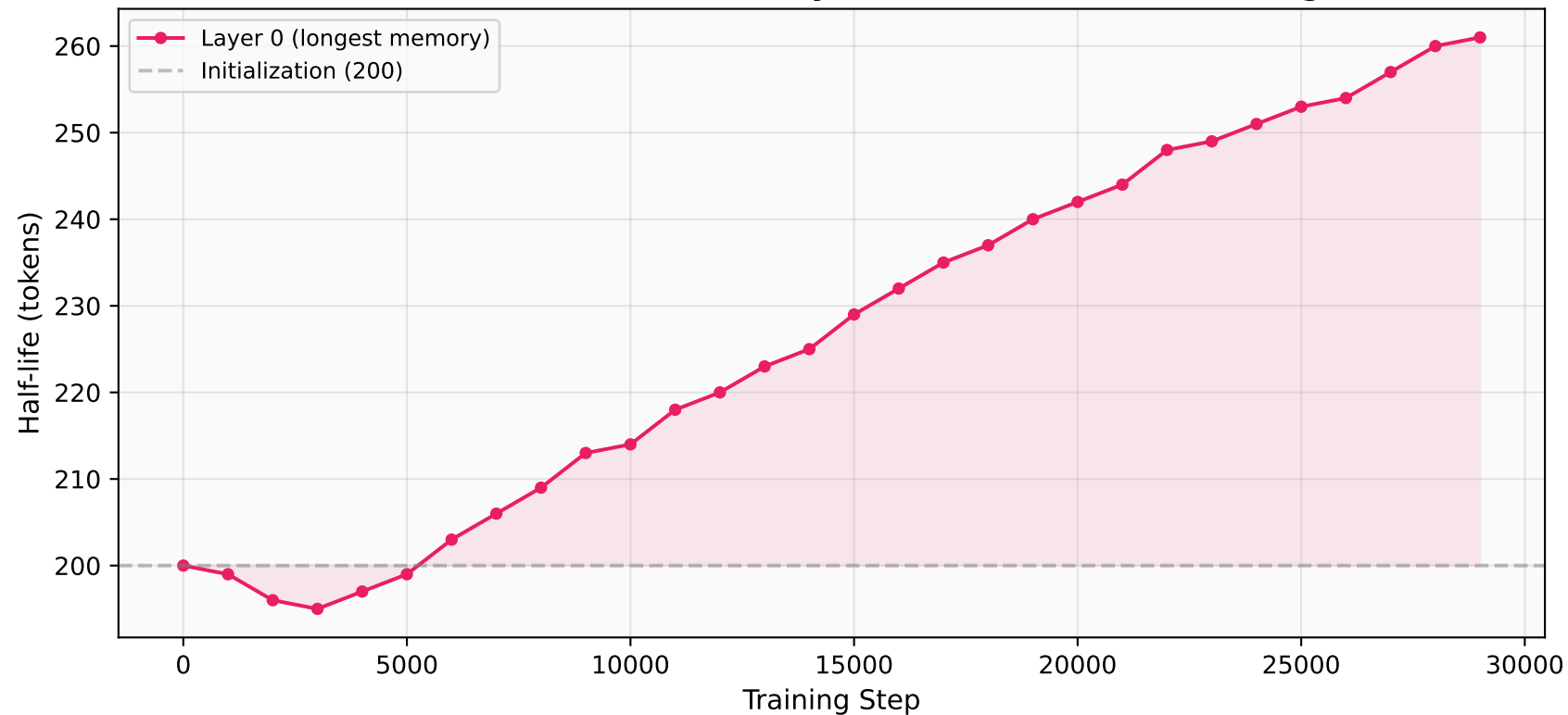
Training Loss



Validation Perplexity

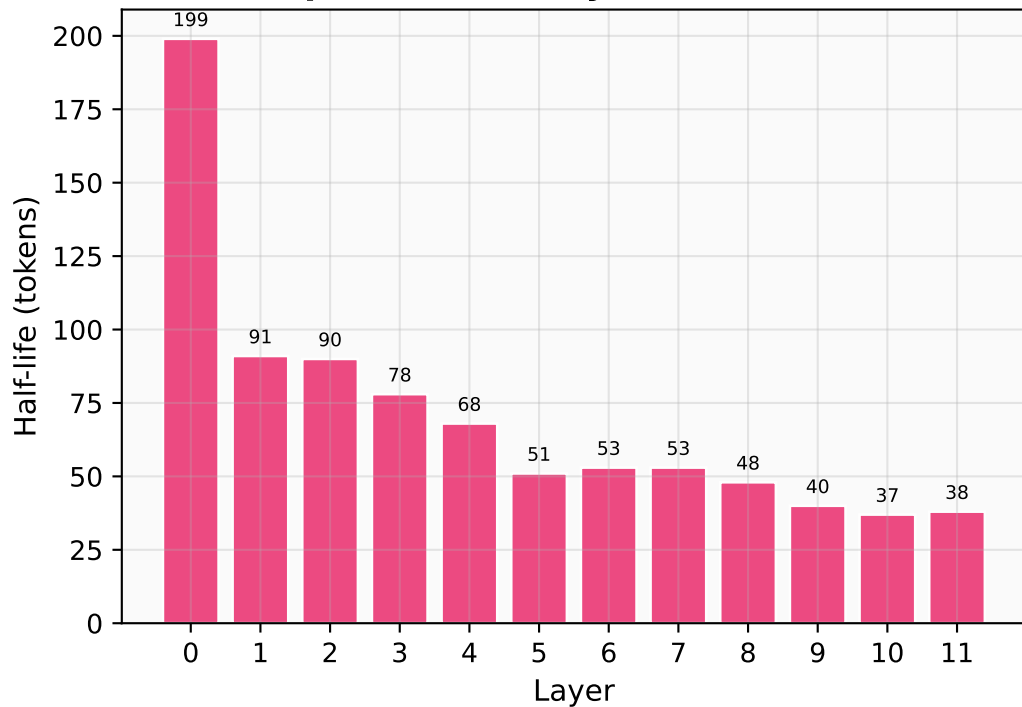


Slow Field Retention — Layer 0 Half-Life Over Training

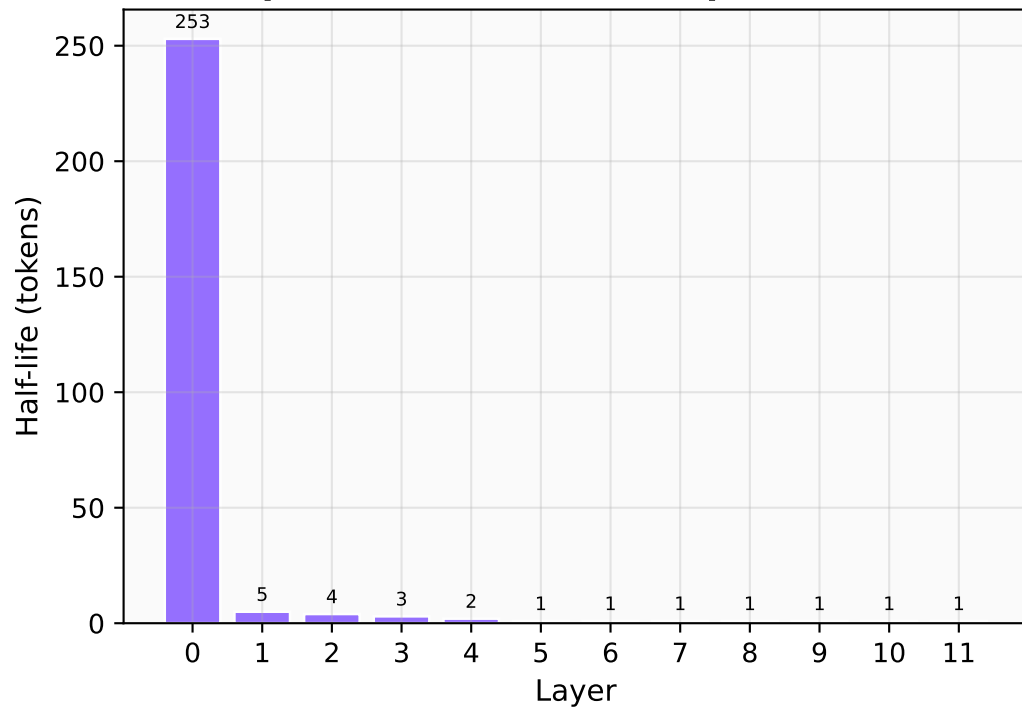


Learned Slow-Field Half-Life by Layer

Step 5,000 — Early Differentiation

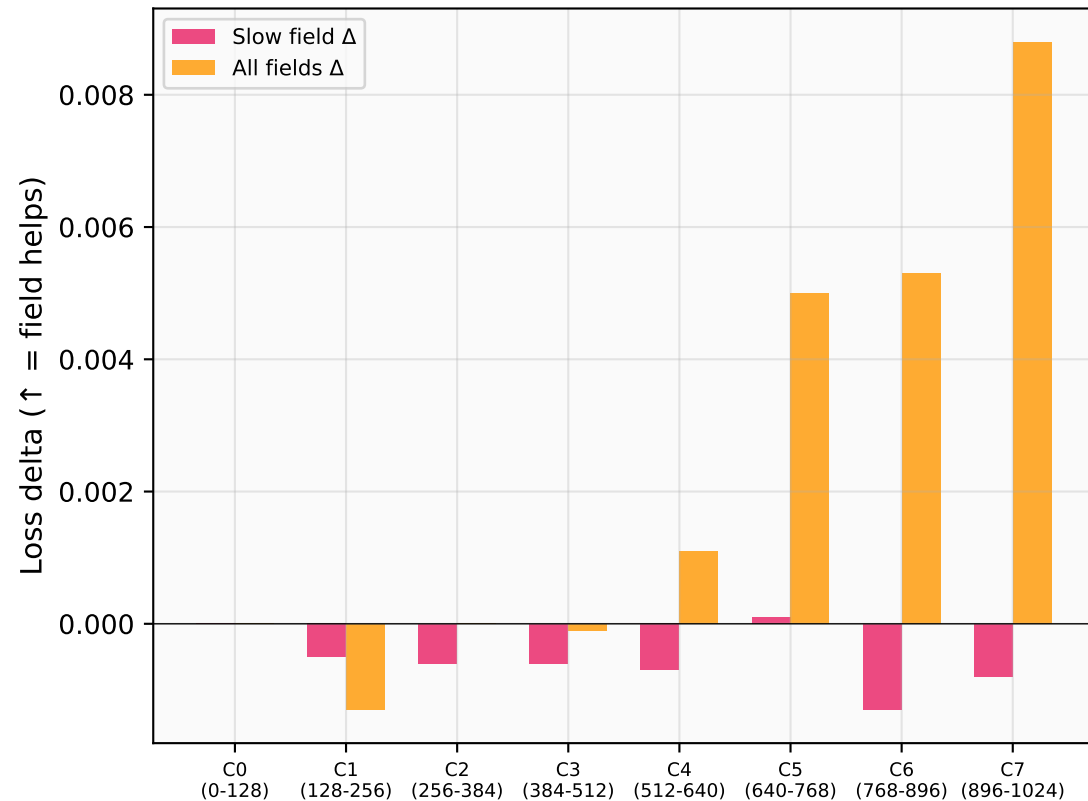


Step 25,000 — Dramatic Specialization

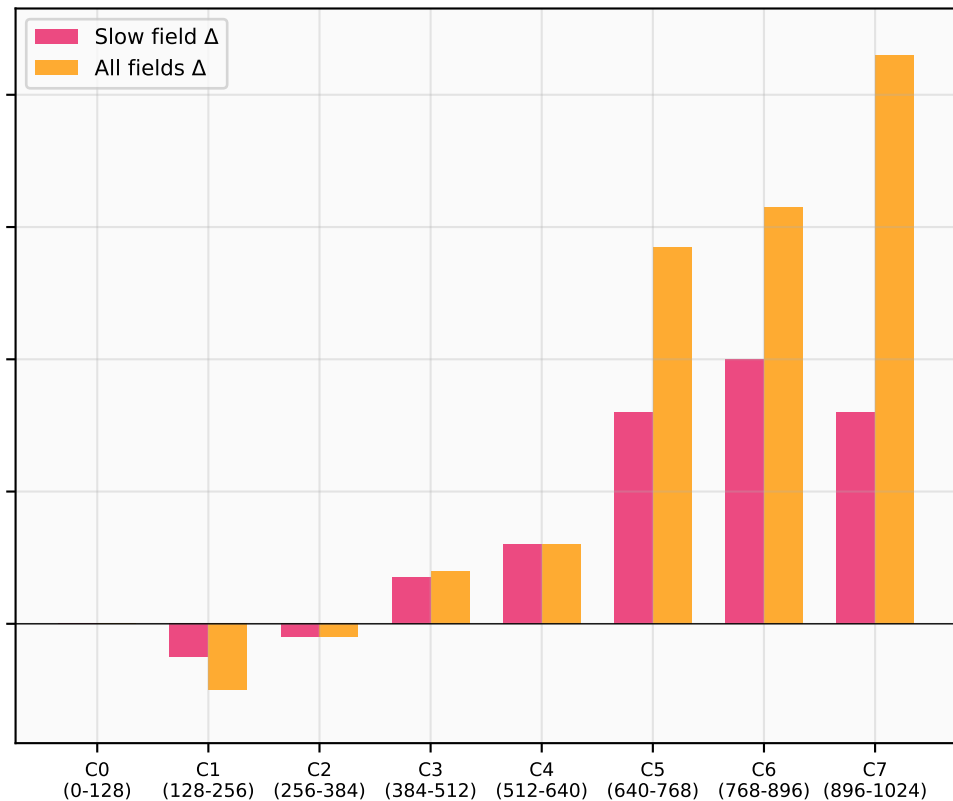


Per-Chunk Ablation: Impact of Removing Field Modulation

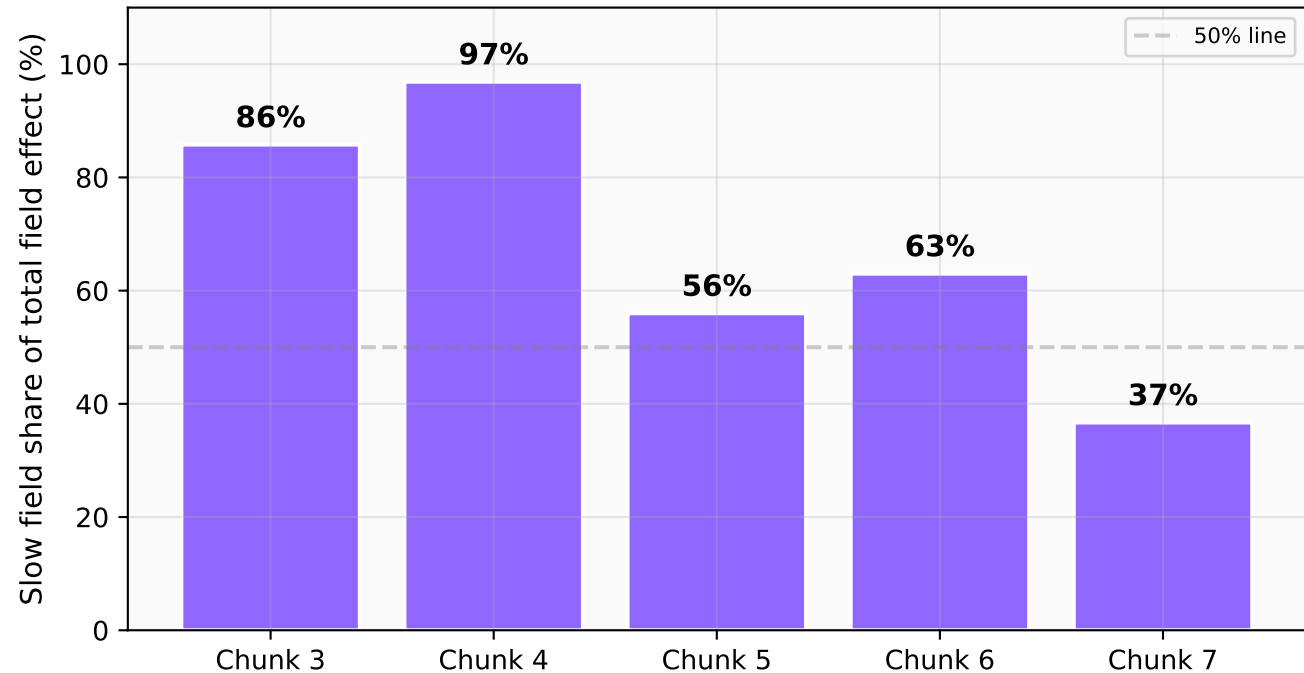
Step 5,000 — Slow field not yet contributing



Step 25,000 — Slow field now active in later chunks



Slow Field Contribution — Step 25,000



Generated Text Samples During Training

Prompt: "The meaning of life is..."

Step 1,000 (PPL 26.7)

The meaning of life is that it is both the process of discovering the true value of life and the process of realizing itself. So I saw the difference in the difference between the two concepts and I think this's why, because it means that the concept of "living" comes from the process of discovering...

Step 5,000 (PPL 30.1)

The meaning of life is that we live in a world where we live in isolation, where we feel we have an obligation to live in a world where we feel that we are alone." The film is a reimagining of contemporary life in the North. The film is about how the British government has been able to control the...

Step 10,000 (PPL 29.4)

The meaning of life is to be alive. I have no idea how to explain it, but once I know what life will do for us, I'll be able to explain it in a way that is realistic. So, for those who might not already know, life is the most beautiful thing. For those who might not know, life is the most realistic...

Step 17,000 (PPL 28.6)

The meaning of life is not only simple but complex. It is also the reason why life is not a random state. It is what it is. It is not a random thing. It is like the Earth, it is what it is. It is not a random thing. It is just an ordinary thing. The Earth has been created by a random person, it is h...

Step 22,000 (PPL 28.0)

The meaning of life is not the point of having to live the life of a man, but a life that is the beginning of a life. The first and only possibility that life and death have for our ancestors. For my father to have the freedom to live in this life and to be there for it to come, was the first and on...

Step 26,000 (PPL 27.5)

The meaning of life is the meaning of freedom. At the end of the book, he describes the life of a man in a suit and tie. The phrase was used to describe the life of a man in a suit and tie. The meaning of life is the meaning of freedom. "There are some that have difficulty with this," he says...

Step 29,000 (PPL 27.3)

The meaning of life is never really clear. The author does not necessarily represent The New American. He is an opinion writer for the magazine. I have been doing this for a while. I've been using the same method for the past few years, but this time I'm using the same method for differ...